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REGULAR PAPER

Real time terrain identification of autonomous robots using machine learning

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Abstract
In this project, machine learning based techniques for real time terrain identification of the autonomous robots are investigated. The factors affecting the performance of autonomous robots include nature of trajectories, on-course obstacles, and nature of terrain. The challenges involved in understanding the terrain of autonomous robots are called localization problems. This project investigates a robust classification based machine learning model to identify the terrains of an autonomous robot from a set of input sensor data , which would incorporated as features in the model. The features are selected with respect to the kinematic and dynamic model of differential drive robots. The terrains are classified into 11 classes and the inputs from different sensors are measured and categorized into the respective classes. A total of 49345 readings were taken. Twenty three classification learning methods are evaluated to find the best fitting model that can identify the terrains of robots in real time. Ensemble Subspace KNN classification learning model produced an accuracy of 100 %, observed as the best model for terrain identification. The results are represented using confusion matrix, which shows the relation between original terrains and model predicted terrains , scatter plot that represents the relationship between each features and ROC Curve analyses each sensor input data. The model output can be provided to an intelligent mechanism to control the wheels of robots and improve their performance.

Keywords Autonomous robots · Terrain identification · Machine learning · Classification learner algorithms

1 Introduction

The developments in the field of robotics have been so instrumental in building a better world to live in. Robotics played a vital role in transforming the world from mechanization to automation. Industry 4.0 is providing highly dynamic changes to the field of automation and robotics. One of the widely researched areas in robotics is the various factors affecting the performance of autonomous robots. Autonomous robots are intelligent robots that can act like humans and take decisions on their own. The development of high performance autonomous robots has contributed in the development of many fields such as industrial, biomedical, agricultural etc.(Kunze et al. 2018). Mobile robots are widely used for autonomous tasks as they can navigate to the desired locations with optimum performance in the time of navigation and the flexibility to use in different locations. Hence, the mobile robots serve as potential candidates for various research activities of autonomous robots. The paper focuses on improving the performance of autonomous mobile robots by analyzing the terrain profiles of its navigation in real-time with the help of a differential drive robot, a common type of mobile robot.

Many researchers investigated the factors affecting the performance of autonomous robots. Along with the trajectory planning, obstacle avoidance and external disturbances are the major factors affecting the performance of robots. Trajectory planning is the primary objective of autonomous robots (Kuleshov et al. 2017; Srebro 2011; Tiep et al. 2018; Pérez et al. 2016; Manko et al. 2015; Altuntas et al. 2016; Samuel et al. 2016; Singh and Bera 2018).Trajectory planning involves steps to reach a particular target, which known as navigation problems. Along with the navigation problems, the robot design is challenging due to some other factors also. Since autonomous robots are not influenced by any external control, they should have the ability to understand the information about environment, especially the existing

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terrain profile. The challenges involved in understanding the terrain of autonomous robots are called localization problems (Tai et al. 2017). Many researchers contributed various methods for identifying the terrain of autonomous robots. The data from internal sensors, which provide feedback will act as the information about the terrain of the system. Garcıa et al. (2008) suggested the use of encoders mounted in each joint, wrist torque sensor and inertial data for the localization. Cheng et al. (2015) suggested a vision-based system which identifies the environment using kinematic model of the robot. Aynaud et al. (2017) suggested an approach, which uses a combination of five lidars, GPS and odometer to analyze the localization. The method of using feedback alone will not be sufficient for terrain identification since the autonomous robots have to decide their action in real-time and the need for advanced techniques are suggested.

Modern technological developments have influenced the research in the field of mobile robots (Mac et al. 2016; Roy et al. 2017; Czarnowskia et al. 2018; Gonzalez et al. 2018; Varghese and Jisha 2018). Researchers have identified that real-time terrain identification of autonomous systems having possibilities of implementing intelligent techniques (Manduchi et al. 2005; Iagnemma and Dubowsky 2004; DuPont et al. 2008). With the developments of machine learning techniques, the pace of the research has increased (Duckworth et al. 2019; Li et al. 2019; Dutta and Dasgupta 2017; Kassahun et al. 2015). Thrun (1998) analyzed the possibilities of machine learning methods in autonomous robot localization. The paper also describes the limitations of conventional feedback algorithms such as lack of flexibility, the dependence of particular types of sensors, lack of optimality, in other words, the lack of knowledge regarding the number of sensors required for data collection and lack of autonomy. They have contributed to the concept of Bayesian Landmark Learning, which extracts data from position sensors and cameras. The Bayesian method was compared with other machine learning methods such as Support Vector Machine (SVM), Optimum Path Forest (OPF), and Multilayer Perceptron with OPF giving accuracy about 98.13% (Almeida et al. 2018). Deep learning-based object recognition techniques are also useful if we use a camera for localization purposes. Kuutti et al. 2018 surveyed the techniques for localization.

The process of self-localization, even though a learning process, also can contribute to the control strategies of autonomous robots. Terrain irregularities will affect the kinematics as well as the dynamics of the robot system. Controllers were previously designed without considering the terrain irregularities or eliminate them by path planning. Robust controllers were developed subjected to speed and acceleration. Adaptive controllers using the dynamic robot models were also designed to compensate for the error due to terrain irregularities. Drews et al. (2019) developed a model predictive control with the help of Deep Convolutional Neural Networks and feature extraction is done using sensors such as camera and wheel speed sensors. Thus incorporating the output of machine learning algorithms to control strategies will be a useful contribution to the performance of autonomous robots. But the researchers face the difficulty of incorporating an accurate terrain identification strategy for control purposes. Along with the accuracy of robots to identify its terrain increases, the performance of control can also be increased.

The accuracy of terrain identification strategies or learning algorithms depends on the feature selections. The features have to be identified from the model of robots. Both kinematic models, which provide information regarding the velocities of mobile robots and dynamic models, provides the forces and torques acting on the robots. Along with the feature selection, the sensors for feature extraction is also important. Giguere et al. (2011) cited the difficulties of using various sensors in obtaining the features. Vision-based systems, radars, etc may not provide a cost-effective solution for the identification of terrains in many cases. The work also points to the need of a robust system for addressing the localization problems. They used a simple tactile probe that is in contact with the environment and its output is trained using classification methods which resulted in an accuracy of 94.6%. The accelerometers which can be placed in the body of robot will be also a tactile probe since it is in contact with the terrain through its wheels. Luo et al. (2017) reviewed various strategies of tactile sensing and evaluated the successful use for various applications of robotics.

As we observed in the literature, a better real-time terrain identification strategy in terms of accuracy and robustness is possible in the case of autonomous robots. A robust method with generalized classes of terrains, irrespective of the application will boost the efficiency of robots. The proposed model identifies 11 different terrains, considering the wheel profiles. This will make our model robust since all possible wheel profiles are discussed and can be used for any application. The paper also investigates the possibilities of incorporating the model in control loops of robots. The paper is arranged as follows: We will present the kinematic and dynamic model of a mobile robot which determines the features to be analyzed. Then the sensors used for data collection are analyzed. Eleven terrain representations which are the categories for learning from the features extracted from sensors, analysis of various machine learning algorithms for the best fitting model is investigated and implementation of the terrain identification model for the control strategies of autonomous robots are discussed in the following sessions.

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2 Kinematic model of the robot

The investigations are carried out using an autonomous mobile robot, NI PITSCO TETRIX which is developed jointly by National Instruments and PITSCO Education. In this robot, two of the wheels are differential driven and the third is a supporting wheel. The position and orientation of the robot is represented in the Fig.1.

Here the value of k, the distance between the target and centroid of the robot is calculated from the coordinates of the robots and is measured from gyroscope sensors. Controller calculates the linear velocity and angular velocity and inverse kinematics techniques will be used for controlling the wheel speed. v and ω or $\dot{\theta}$ refer to the linear and angular velocities of the robot. The forward kinematics is given as

$$v = \frac{d}{4}(\omega_{rightwheel} + \omega_{leftwheel}) \tag{1}$$

$$\dot{\theta} = \frac{d}{2l}(\omega_{rightwheel} + \omega_{leftwheel}) \tag{2}$$

Using inverse kinematics analysis the angular velocities of the left and right wheel, which are derived from the equations of forward kinematics are calculated as follows:

$$\omega_{leftwheel} = \frac{1}{2d}(v - \frac{\omega}{2}) \tag{3}$$

$$\omega_{rightwheel} = \frac{1}{2d}(v + \frac{\omega}{2}) \tag{4}$$

The angular velocities will hence determine the navigation of the robots. The values of angular velocities of both the wheels depend on the robot parameters.

3 Dynamic model of the robot

The dynamic model of a differential drive robot gives the information about the forces and torques acting on the system. The dynamics of a mobile robot is non linear due to its non holonomic constraints. Linearization of the model is done by reducing the number of holonomic constraints from n to m such that m < n, where n is the number of kinematic constraints.Now we can apply Lagrangian function for robotic movement

$$\mathcal{L} = \mathcal{K} - \mathcal{P} \tag{5}$$

where $\mathcal{K}$ is the kinetic energy and $\mathcal{P}$ is the potential energy.

Kinetic energy can be represented by

$$\mathcal{K} = \frac{1}{2}\dot{\theta}^T \mathbf{B}\dot{\theta} \tag{6}$$

where $\mathbf{B}$ is the inertia matrix of the mobile robot coordinates. Thus equation(5) can be written as:

$$\mathcal{L} = \frac{1}{2}\dot{\theta}^T \mathbf{B}\dot{\theta} - \mathcal{P}(\theta) \tag{7}$$

The Lagrnages equation can be written in the form

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right)^T - \left(\frac{\partial \mathcal{L}}{\partial \theta}\right)^T = \mathbf{J}(\theta)\tau + \mathbf{A}(\theta)\lambda \tag{8}$$

where $\mathbf{J}(\theta)$ is nxk matrix, k=n-m, this is in assumption that system is linear.$\mathbf{A}(\theta)$ is the matrix representing kinematic constraints and λ is the lagrange multiplier. We can represent the $\mathbf{A}(\theta)$ in Pfaffian form as

$$\mathbf{A^T}(\theta)\dot{\theta} = 0 \tag{9}$$

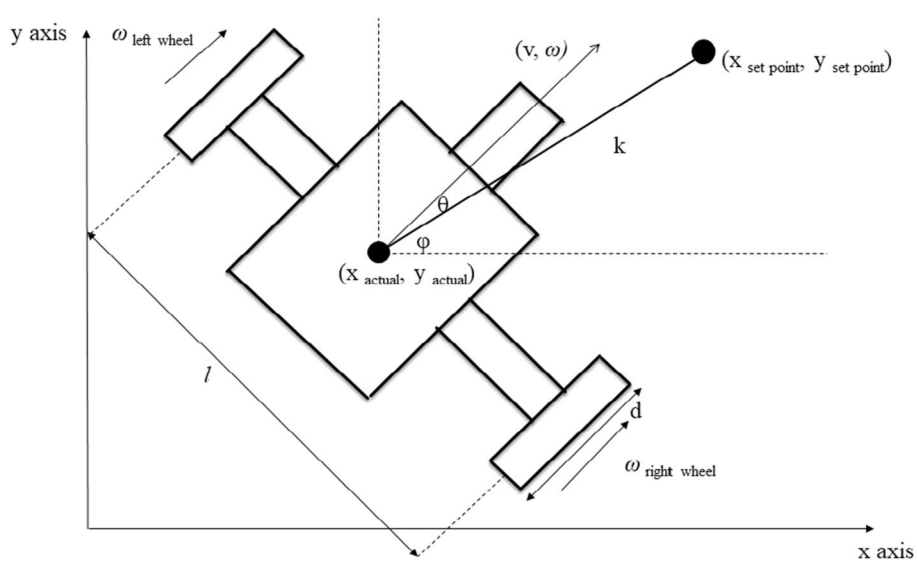

Fig. 1 Model of two wheel differential drive robot

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Substituting the Lagrangian function, the equation will be in the form

$$\mathbf{B}\ddot{\theta} + \mathcal{Z}(\theta, \dot{\theta}) = \mathbf{J}(\theta)\tau + \mathbf{A}(\theta)\lambda \tag{10}$$

where

$$\mathcal{Z}(\theta, \dot{\theta}) = \dot{\mathbf{B}}\dot{\theta} - \frac{1}{2}\left(\frac{\partial}{\partial\theta}(\dot{\theta}^T\mathbf{B}\dot{\theta})\right)^T + \left(\frac{\partial\mathcal{P}}{\partial\theta}\right)^T \tag{11}$$

The Lagrangian multipliers are unknown values and hence these can be replaced by introducing $\mathbf{R}$ such that

$$\dot{\theta} = \mathbf{R}\eta \tag{12}$$

$$\dot{\eta} = \left[\dot{\theta}_R \dot{\theta}_L\right]^T \tag{13}$$

$\mathbf{R}$ is the null space of the matrix $\mathbf{A}(\theta)$.i.e.

$$\mathbf{R}^{\mathbf{T}} A^T(\theta) = 0 \tag{14}$$

Taking derivative of (12),

$$\ddot{\theta} = \dot{\mathbf{R}}\eta + \mathbf{R}\dot{\eta} \tag{15}$$

Substituting equation(15) in equation (10),

$$\mathbf{B}(\dot{\mathbf{R}}\eta + \mathbf{R}\dot{\eta}) + \mathcal{Z}(\theta, \dot{\theta}) = \mathbf{J}(\theta)\tau + \mathbf{A}(\theta)\lambda \tag{16}$$

This can be reduced to the form

$$\bar{\mathbf{B}}\dot{\eta} + \bar{\mathcal{Z}}(\theta, \dot{\theta}) = \bar{\mathbf{J}}(\theta)\tau \tag{17}$$

where

$$\bar{\mathbf{B}} = \mathbf{R}^{\mathbf{T}}\mathbf{B}\mathbf{R} \tag{18}$$

$$\bar{\mathcal{Z}} = \mathbf{R}^{\mathbf{T}}\mathbf{B}\dot{\mathbf{R}} + \mathbf{R}^{\mathbf{T}}\mathcal{Z}(\theta, \dot{\theta})\mathbf{R} \tag{19}$$

$$\bar{\mathbf{J}} = \mathbf{R}^{\mathbf{T}}\mathbf{J}(\theta) \tag{20}$$

Equation (17) represents the reduced dynamic model of the Differential Drive Robot. By observing the kinematic and dynamic models of robots, the features should represent the linear position,velocity, acceleration as well as the angular position, velocity and acceleration. This can be analyzed using following sensors as given in the next session.

4 Instrumentation for data collection

The sensors used are gyroscope, orientation, gravity and rotation sensors. Acceleration along x, y, z axes represented by accelerometer, force of gravity along x, y, z axes represented by gravity, rate of rotation along x, y, z axes represented by gyroscope, rotation vector components along x, y, z and scalar rotation component represented by rotation sensors, orientation along x, y, z is calculated by orientation vector.

5 Terrain representations

The process of defining the terrains is explained here. The categorical variables are defined according to the position of both wheels, which can represent the terrains. Table 1 shows the classification of wheel profiles and details are explained below.

1. Rest (T1)
   The state of rest is termed as T1, the algorithm can give the categorical output as "Rest". There will not be any disturbance due to the terrain if the mobile robot continue in its state.
2. Simple straight run (T2)
   In the second scenario, the robot moves in a simple flat surface with or without turning. Here the signals to servo motors which controls the wheel rotation will not be much regulated by terrain profile. The robot may move in a straight line or turning surface as per target identification. Simple straight run occurs when the robot is moving through the flat surface
3. Moving up a slanting surface (T3)
   The effect of terrain profile change on the path of autonomous mobile robots is analyzed with . If the robot design is based on target tracking and trajectory planning, it may fail to intelligently recognize its moving terrain profile. But an intelligent system must be able to understand the profile of its navigation and both wheels should be controlled accordingly.
   As described in our block diagram the terrain profile information will give as the correction factor through a sensible mechanism. When a robot moving to an upper surface through slanting surface, the traction will affected by the frictional force, gravitational force etc of the surface. This profile is defined as type 3.
4. Moving down a slanting surface (T4)
   This is termed as Type 4 terrain profile, where the inverse of type 3 occurs. Effect of traction on the both wheels are same as in the previous case. The information regarding this profile can able to give a tighter control to both wheels in same amplitude, which could opposes the forces acting on it. This information can be used also in calculating the time required for the navigation of the autonomous system.
5. Moving on a surface with right wheel up or moving on a surface with left wheel up (T5, T6)
   These profiles are termed as Type 5 and Type 6 in our analysis. The difference from the previous profiles is one of the wheel is moving through a secondary surface which

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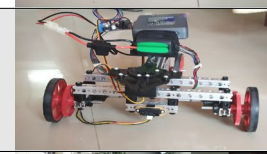
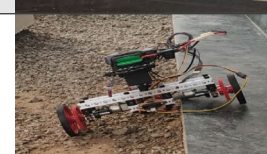
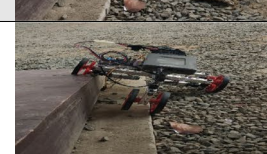
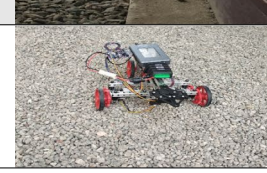
Table 1 Terrain representations

Sl.No	Robot orientation	Type	Image
1	Rest	T1	
2	Simple straight run	T2	
3	Moving up a slanting surface	T3	
4	Moving down a slanting surface	T4	
5	Moving on a surface with right wheel up	T5	
6	Moving on a surface with left wheel up	T6	
7	Moving up a slanting surface with right wheel up	T7	
8	Moving up a slanting surface with left wheel up	T8	
9	Moving down a slanting surface with right wheel up	T9	
10	Moving down a slanting surface with left wheel up	T10	
11	Continuous rough surface	T11	

is inclined. The sample sensor values are given in the table. Here the output from learning algorithm will control a select mechanism to control the PWM signal and angular velocities of both wheels ($\omega_{leftwheel}, \omega_{rightwheel}$).

6. Moving up a slanting surface with right wheel up or left wheel up (T7, T8).

The Type 7 and Type 8 categories are analyzed with respect to these constraints. So the logic mechanism

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should select a velocity profile for the left wheel and right wheel that should compensate the effects of frictional force as well as forces opposing the balance.

7. Moving down a slanting surface with right wheel down or left wheel down (T9, T10).

   These are categorized into Type 9 and Type 10 which is the inverse of Type 7 and Type 8 respectively. These categories require control signals according to the intelligent mechanism, which generates wheel profiles using the predicted variable from classification learner output.

8. Continuous rough surface (T11).

   The rough surface is defined under Type 11. The understanding about these profiles will be helpful in controlling the angular velocities to a controlled limit and hence can increase the balance of the system. The time calculations are also computed accordingly.

6 Classification learning on a data set of 11 categorized terrains

Overall, 49345 readings are measured. A 49345×18 matrix is used for training. Out of the 18 columns, 17 represent the sensor data or features and last column represents the categorical variable, terrain . The 17 feature variables include x, y, z coordinates from accelerometer, representing the acceleration; three-dimensional outs from gyroscope, orientation sensors, gravity and rotation sensors. The 49345 rows represent the measurements taken from 11 types as represented in Table 1.

Different machine learning concepts are analyzed separately to implement a suitable control strategy. The classification algorithms which use the fundamental principles of the methods such as Decision Trees, Discriminants, Naïve Bayes, Support Vector Machines methods (SVMs), k-Nearest Neighbor Algorithm (KNN), Ensembles methods, etc. are examined for the purpose. This is the major step of our analysis, because a suitable machine learning method in terms of efficiency, accuracy, speed will have to be identified for the control mechanism of mobile robots since software output will be pertinent to the real-time process.

The concept of Decision Tree method fundamentally generates a tree-like structure of decisions arranged from top to bottom and the logical reasoning is practiced based on the observation of feature values in separating the different classes, in our case the terrain categories. Considering the number of observations, the reduction of variance concept is used for selecting the best sensor data that can separate the terrain classes. For example, to separate between the classes "Rest" and "Simple straight run", the gravity sensor value along the z-axis has the least variance with a value of 6.07, in rest and 0.000316 in a straight run. Also the range of the same parameter in the range of 9.79300–9.79303 and that of 9.79200– 9.792305. The variance value is taken along the accuracy using the Gini Index is also analyzed. Gini Index is the probability calculation between each class. By combining these, the concept of the decision tree is implemented by creating logical reasoning regarding the class of the terrain according to the numerical values specified in the previous step. Here we analyze three separate trees; Coarse Tree, with maximum 4 splits (split means the number of branches from each node, which will be the four expected possible outcomes of the problem) gives an accuracy of 53.9%, Medium Tree with 20 splits with an accuracy of 79.2%, and Fine Tree with maximum 100 splits is having an accuracy of 89.4%.

The principles of Discriminant Analysis are evaluated by considering two separate methods; Linear Discriminant Analysis and Quadratic Discriminant Analysis. The Linear Discernment Analysis(LDA) is analyzed by forming a linear relationship between the features and the categories. The primary challenge is to identify the most suitable feature to represent the categories with a linear function. The Eigenvalues of the scatter matrices within and between the classes are used for creating new features for linearizing the relationship with the categories. Quadratic Discriminant Analysis, on the other hand, is a type of Linear Discernment Analysis which will not reduce the dimension of the features, the covariance of the matrix is analyzed each time. The accuracy of the Linear Discriminant method is 71.3% and the Quadratic Discriminant is 84.2%.

Naïve Bayes techniques use the independence of each feature. The probability of occurring a terrain class is analyzed based on the posterior probability of the features. Hence the output model will depend upon the level of independence of the features. There are two different types of Naïve Bayes is analyzed in this paper; Gaussian Naïve Bayes and Kernel Naïve Bayes. The classification is based on the function to select the boundaries of the feature data set. Kernel Bayes classifier gives an accuracy of 84.5% where Gaussian Naïve Bayes gives an accuracy of 77.8% with a training time of 7.2301 s. Support Vector Machines (SVM) are another important type of supervisory learning algorithm used in classification problems. SVMs normally creates a hyper line to differentiate the feature values of different classes. For eg; to classify between the state of rest or simple run, we can draw a line in between the cluster of feature values. The data points near to this line are known as support vectors. The gap between the support vectors determines the accuracy of the model, which is called the margin of SVMs. There are different types of approaches to SVMs in machine learning such as Linear SVMs, Quadratic SVMs, and Gaussian SVMs. This classification is based on the selection of a hyperplane to classify the vectors. Linear SVMs will give an accuracy of 84.5% accuracy with a training time of 402.52 s. The Quadratic SVM will give an accuracy

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of 94.7% with a training time of 500.71 s. The Gaussian function SVM is classified into three; Fine Gaussian, Medium Gaussian, and Coarse Gaussian. Fine Gaussian gives an accuracy of 97%, Medium Gaussian gives an accuracy of 97% and Coarse Gaussian gives an accuracy of 85.77%.

The analysis of classification learning is extended to another popular algorithm, the k-Nearest Neighbor Algorithm. (KNN). It is one of the simplest algorithms. The algorithm will analyze "k" number of neighbors that features and adapts the same category that the most features in the neighborhood represent. For example if k=1, the output of the model will be the category of the next nearest feature represents. Thus the selection of k value is very important in determining the accuracy of the system. The KNN approaches can be classified into different types based on the number of k and the method of selecting the boundaries. If k=1, the method is fine KNN and gives an accuracy of 99.5 % with a training time of 167 s. For k=10, the method is termed as medium KNN and gives an accuracy of 98.1 %. With a training time of 108.1 s. Coarse KNN is the method with k=100; it provides an accuracy of 91.7% with a training time of 35.5 s. In all the above methods Euclidean Distance is measured for selecting k. There are other two methods, cosine KNN and Cubic KNN where the k is selected based on Cosine Distance and cubic area respectively. Cosine KNN gives an accuracy of 98.2% with a training time of 114.12 s, cubic KNN with an accuracy of 97.3% with a training time of 347 s. Another method called weighted KNN is used where weights are assigned to neighbors with the nearest neighbor has more weight that farther neighbor. Hence we can increase the number of k without affecting the optimality. This has resulted in an accuracy of 99% with a training time of 114 s.

The possibilities of combining different algorithms are tested using the Ensembles method. Ensembles of various decision tree algorithms are analyzed with Ensemble-Boosted Tree gives an accuracy of 80.4% with 185.02 s, and Ensemble-Bagged Tree gives an accuracy of 99.9% with 97 s of training time. The Ensemble-Discriminant give an accuracy of 73% with 44.7 s training time and Ensemble- RUSBoosted Tree gives an accuracy of 80.6 percent within a training time of 70 s. Interestingly, the method of Ensemble-Subspace KNN successfully identified the categories with an accuracy of 100 % in training time of 46.379 s. This method uses subspace for finding the value of k. The Table 2 summarizes results of all the methods with model accuracy, training time and prediction speed of the model, represented in the number of observations per second.

7 Experimentation results

Since the predicted variable is a categorical one, classifier learner of machine learning toolbox in MATLAB is used. Under the 22 different methods considered, Ensemble-Subspace KNN give the best result with a predicted speed of 6600 obs /sec and training time is 53.08 s. The number of learners for ensembles is 30 and subspace dimension is 9. The cross validation technique is used so that the data for training and validation are not provided separately. The confusion matrix, ROC Curve and Scatter plot of the model is represented in Figs. 2, 3 and 4 respectively.

Confusion matrix represents the briefing of the prediction results of classification learner. In our case the confusion matrix is a 11x11 matrix with predicted class in the x axis and true class in the y axis. Confusion matrix not only represents the prediction accuracy, but also some mathematical calculations that represent parameters that can act as quality indicator to our predicted model.

The variables are classified into following four domains:

True positive (X):- If the observed class of terrain is positive and predicted class is negative.
False negative(Y):- If the observed class of terrain is positive and predicted class is negative.
True negative (C): If the observed class of terrain is negative and predicted class is negative.
False positive (D): If the observed class of terrain is negative and predicted class is positive. The accuracy can be calculated as follows:

$$\frac{(X+C)}{(X+Y+C+D)} * 100 \tag{21}$$

Fig. 3 represents the accuracy of the confusion matrix in each class. The accuracy of seven of the eleven classes touches 100% and remaining classes have an accuracy of around 99%. In classification learner algorithms, the number of correct positive predicted terrain values to the total number of actual terrain positive values.

$$Recall = \frac{X}{(X+Y)} * 100 \tag{22}$$

The term precision represents the correct positive predicted terrain values to the total number of positive predicted terrain values.

$$Precision = \frac{X}{(X+D)} * 100 \tag{23}$$

The recall and precision values are together known as Information Retrieval Ratio (IRR). The value of IRR is 100% in the simulation. Both the precision and recall values can be used to find the value of F-measure which is

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Table 2 Analysis of learning methods for terrain identification

Sl.No.	Learning algorithm	Training accuracy (%)	Training time (sec)	Prediction speed (obs/sec)
1	Fine Tree	89.4	8.1883	290000
2	Medium Tree	79.2	6.7015	370000
3	Coarse Tree	53.9	5.0465	350000
4	LDA	71.3	14.6365	120000
5	Quadratic Discriminant	84.2	84.2	150000
6	Kernel Naïve Bayes	84.5	7.2301	96000
7	Gausian Naïve Bayes	77.8	2193	76
8	Linear SVM	84.5	402.52	18000
9	Quadratic SVM	94.7	500.71	6300
10	Cubic SVM	96.3	1020.6	6700
11	Fine Gausian SVM	98.7	1860	240
12	Medium Gausian SVM	97	345.92	1900
13	Coarse Gaussian SVM	85.77	471.83	980
14	Fine KNN	99.5	167	15000
15	Medium KNN	98.1	108.8	16000
16	Coarse KNN	91.7	35.5	11000
17	Cosine KNN	98.2	114.12	14000
18	Cubic KNN	97.3	3474	47
19	Weighted KNN	99	114	14000
20	Ensemble-Boosted Trees	80.4	185.52	35000
21	Ensemble-Bagged Trees	99.9	97.13	29000
22	Ensemble-Subspace Discriminant	73	44.76	9000
23	Ensemble-RUSBoosted Trees	80.6	69.378	40000
24	Ensemble-Subspace KNN	100	46.37	7300

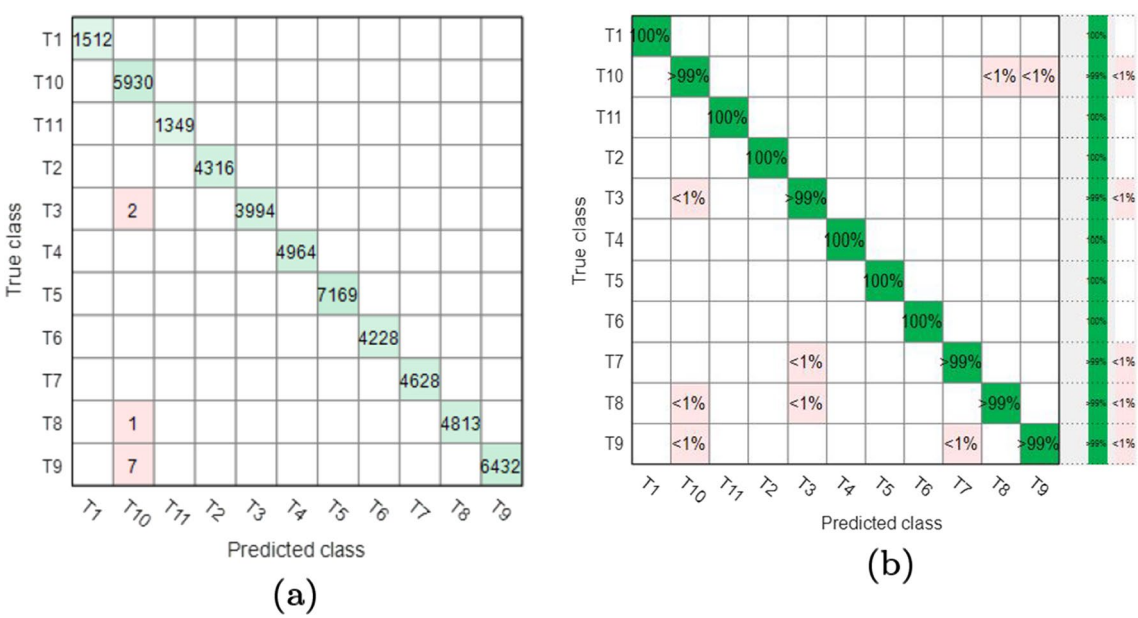

Fig. 2 **a** Relationship between true class and predicted class with respect to number of observations. **b** Relationship between true class and predicted class with respect to percentage of accuracy

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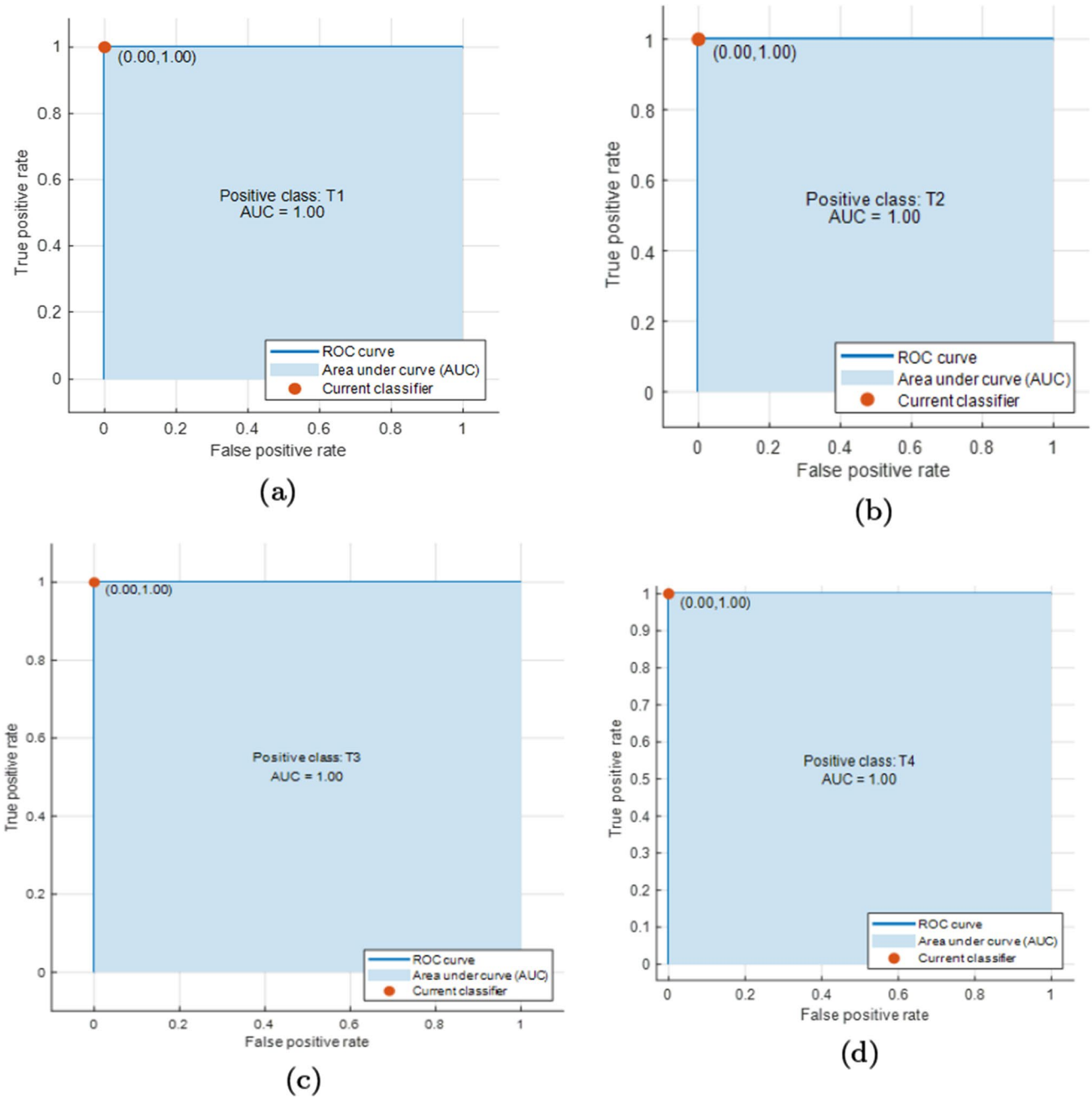

Fig. 3 Examples of ROC Curve representing the relation between true positive rate and false positive rate of predicted terrains **a** of T1 **b** of T2 **c** of T3 **d** of T4

the indication of the ability of the learner to retrieve the information. The value of F- Measure is also 100 %.

The Receiver Operating Characteristics (ROC) Curve analyses the performance of the classification learner using area under the curve. The area depends on two factors; sensitivity and specificity. The sensitivity represents the ratio of true positive classes to the sum of true predicted class values. This is represented in y axis as true positive rate. False positive rate is represented in x axis, which is the inverse of specificity. Specificity is the value of true negative classes ratio to the total number of true predicted values. The examples of ROC Curve for four categories are shown in Fig. 3. A square shaped ROC Curve is pointing to the optimum performance of the developed model.

The scatter plot represents the interrelationship among the features and the ability of features to categorize each terrain. All the 136 combinations of the 17 input sensors are being analyzed using scatter plot. Examples of this analysis

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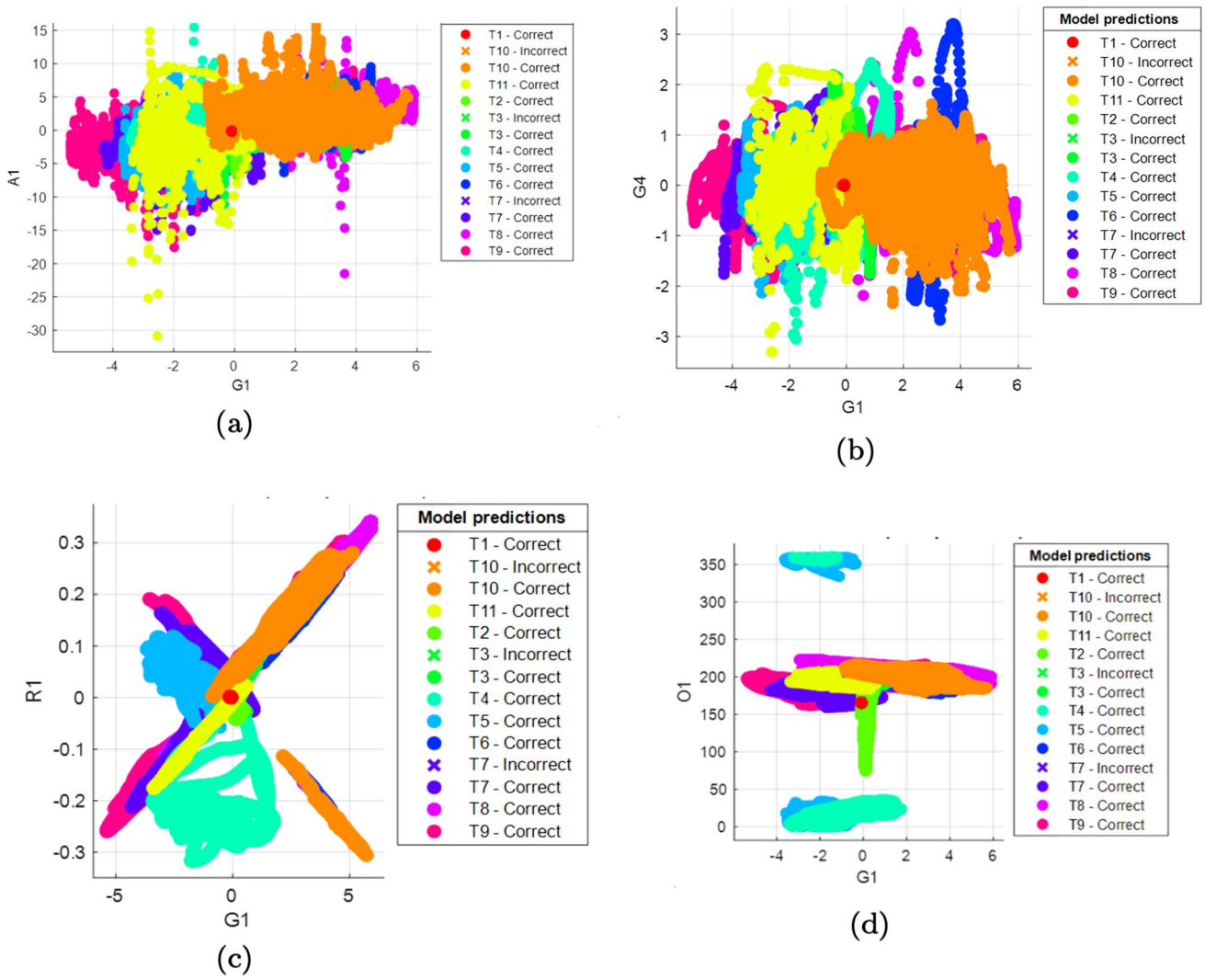

Fig. 4 Examples of scatter plot representing the relationship between sensor data, **a** Between gravity along x axis (G1) and acceleration along x axis (A1). **b** Between gravity along x axis (G1) and rate of rotation about x axis (G4). **c** Between gravity along x axis(G1) and rotation vector along x axis(R1). **d** Between gravity along x axis(G1) and orientation along x axis(O1)

are shown in Fig. 4. The distribution of parameters for each terrains is analyzed and the ability of the scatter plot to distinguish the types of terrains is listed. The distribution of scatter plot is the indication of the accuracy of the model.

8 Implementation of terrain identification model in autonomous robots

The block diagram for controlling the mobile robots is shown in Fig. 5. The setpoint for the loop is the pose of the target that is a position, and orientation ($x_{setpoint}, y_{setpoint}, \theta_{setpoint}$). The robot navigates to the corresponding position, the desired trajectory with the help of a control mechanism. A trajectory controller compares the current pose of the robot with the desired values, with the help of a comparator. The outputs of the trajectory controller are used for further control (P_x, P_y, P_θ). The inverse kinematics block calculates the linear and angular velocities of the robot and the angular velocities to the left and right wheel using equations (3 and 4) . There are disturbances in the control mechanism developed from the nature of terrains of robot navigation.This can be compensated by driving the two wheels at separate velocities, according to the type of terrains. The process will help in the smooth motion of the robot and improve the performance during braking and sudden stoppage. In differential drive robots, though two wheels are separately driven with two DC motors, a conventional controller may not able to produce the desired speed for each wheel that can compensate the terrain effects. An intelligent control mechanism with the help of the proposed machine learning model can be used to select the velocities to the left and right wheel. The

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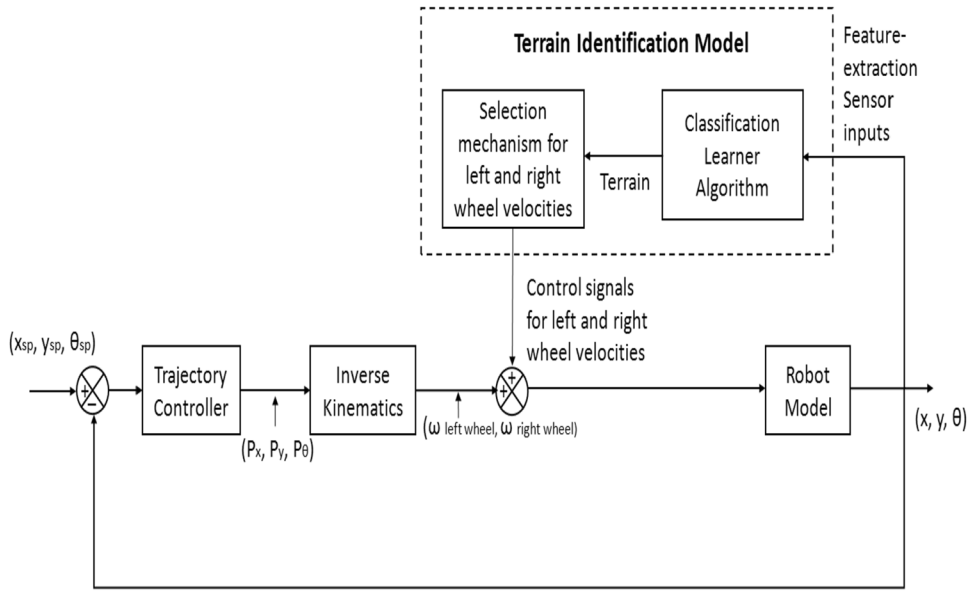

Fig. 5 Block diagram of robot navigation system with terrain identification model

terrains are identified in terms of the wheel profiles and this simplifies the process of choosing the optimum velocities. The model can be implemented with feedback from the robot model. The feedback from the robot to the machine learning model consists of the linear position, velocity, acceleration as well as angular position, velocity and acceleration. The sensors explained in session 4 are measuring the inputs to the model. The inputs are the features of the proposed model. The output of the classification learner algorithm is the category of terrain identified by the machine learning model. The autonomous robots can act like humans in perception of its surroundings. The future scope of the project is to develop a logical algorithm for selecting the velocity profile of both the right and left wheels, corresponding to each terrain. The control variable from the model is added along with wheel velocities from the trajectory controller. Hence the robot can compensate the errors occurring due to the terrain irregularities. The process has to be calibrated for the optimum performance of the robot.

9 Conclusion

The project focuses on real-time terrain identification of autonomous robots. The purpose of the project is to improve the human like performance of autonomous robots in identifying its existing terrain by perception, and to compensate the errors developing due to it. The objective has been achieved by the classification learner model using the Ensemble-Subspace KNN method. The model gives an accuracy of 100 % in identifying the terrain. In other words, the localization problem is addressed with a best-fitting machine learning model. The model is robust and can be implemented irrespective of the application of the robot. This model is added to the trajectory planning and control of a mobile robot. In this model, the terrain profiles are identified using 17 features, which are data from different sensors. The features are chosen considering the kinematic and dynamic model of the robot. A data set with 49345 readings are analyzed with 11 categorical variables as output representing the terrains. 23 classification algorithms are analyzed using MATLAB and Ensemble-Subspace KNN is identified as the best-fit algorithm. Analysis of confusion matrix, scatter plot and ROC curve are carried out for analysis. The results are investigated to identify the relationship between various sensor values. All these data are presented. The concept of an intelligent system that can use the terrain identification model for helping the control of autonomous robots is presented. The future scope of the work is in developing a control algorithm for selecting the wheel velocities according to each terrain identified by the Ensemble-Subspace KNN model and thereby improving the performance of robots.

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